

CLOUD-BASED PHARMACY MANAGEMENT SYSTEM

Final Project Report

Project Team:

Andrea Kitanovska (89231153)
Samoil Gjorgiev (89231015)
Ana Milenkoska (89231022)
Hristijan Kochovski (89231045)

Table of Contents

Table of Contents.....	1
Chapter 1: Project Charter.....	3
1.1 Project Identification.....	3
1.2 Project Client.....	4
1.3 Purpose and Objectives.....	4
1.4 Importance of the Project.....	4
1.5 Technologies and Tools.....	5
1.6 Description of the finished product, service or result.....	5
1.7 Project Scope and Limitations.....	5
1.8 Project Schedule and Milestones.....	5
1.9 Project Team.....	5
1.10 Risk Assessment.....	6
Chapter 2: User Research — Personas and Problem Scenarios.....	7
2.1 Persona 1: Dr. Vangel Markovski — General Practitioner.....	7
2.2 Persona 2: Pharmacist Lea Zajkov — Private Pharmacy Chain.....	8
2.3 Persona 3: Elena Petrova — Foreign University Student.....	9
Chapter 3: User Stories and Acceptance Criteria.....	11
3.1 Doctor User Stories.....	11
3.2 Pharmacist User Stories.....	11
3.3 Patient (International Student) User Stories.....	12
Chapter 4: Hypothesis Testing and User Interviews.....	14
4.1 Research Hypotheses.....	14
4.2 Interview Findings.....	14
4.3 Key Findings and Identified Improvements.....	15
Chapter 5: Usability Testing.....	16
5.1 Testing Type and Goal.....	16
5.2 Participants.....	16
5.3 Session Structure.....	16
5.4 Tasks.....	16
5.5 Task Completion Times.....	17
5.6 Key Findings and Recommended Improvements.....	17
Chapter 6: Agile Development — Scrum Sprints.....	18
6.1 Sprint 0: Project Setup and Planning.....	18
6.2 Sprint 1: Personas, Scenarios, and User Stories.....	18
6.3 Sprint 2: Validation and Usability Testing.....	19
6.4 Sprint 3: Prototype Improvement and System Implementation.....	19
Chapter 7: Project Stakeholder Analysis.....	21
7.1 Pharmacists.....	21
7.2 Suppliers and Medicine Distributors.....	21
7.3 Health Authorities — Ministry of Health.....	21
7.4 Patients.....	22
Chapter 8: Engineering Management — Patterns and Practices.....	23

8.1 Negative Pattern: Ignore Human Issues.....	23
8.2 Negative Pattern: Last-Minute Development.....	23
8.3 Positive Pattern: Continuous Integration and Testing.....	24
8.4 Positive Pattern: Visibility of System Status.....	24
Conclusions.....	26

Chapter 1: Project Charter

This chapter presents the foundational document of the project, establishing its scope, objectives, team composition, risk assessment and contractual basis. It serves as the formal agreement between the development team and the client.

1.1 Project Identification

Project Code	CBPMS-2026
Project Name	Cloud-Based Pharmacy Management System
Version	1.0
Date	21 February 2026
GitHub	https://github.com/sam0il/Cloud_Based_Pharmacy_Managment_System

1.2 Project Client

The client for this project is the Healthcare System of the Republic of Slovenia. The designated representative is Trpe Markovski, IT Manager in the Ministry of Health.

1.3 Purpose and Objectives

The purpose of the Cloud-Based Pharmacy Management System is to ensure that patient prescriptions are always available digitally, regardless of which hospital or pharmacy the patient visits. The system is also designed to reduce paperwork, accelerate prescription processing and improve overall system safety through automated alerts and centralized data management.

1.3.1 Key Objectives

- Allow pharmacists to search patients, view medical records and manage medication inventory.
- Automatically update stock levels after each sale or prescription fulfillment.
- Send alerts when medicines are running low or nearing their expiry date.
- Support the digital processing of electronic prescriptions issued by doctors.
- Automatically link prescriptions to the patient's medical record.
- Provide clear reports on sales, stock status and suppliers.
- Enable data synchronization between different pharmacy branches.
- Allow doctors to add, search and manage patient records.
- Allow administrators to manage user roles and permissions across the system.
- Ensure all data is stored securely and regularly backed up.
- Support real-time data access across all pharmacy branches.
- Support role-based access control for Doctor, Pharmacist and Admin roles.

1.4 Importance of the Project

The Cloud-Based Pharmacy Management System is important because it replaces outdated and disconnected pharmacy software with one centralized platform. This will improve inventory tracking, prescription processing and sales management, reducing errors and saving time. The system will also increase data security through controlled access, secure storage and backups and help the pharmacy meet healthcare regulations. In addition, it will improve customer service and patient safety by enabling faster prescription handling and better cooperation with doctors and healthcare systems. Overall, the project improves efficiency, security and quality of pharmacy services.

1.5 Technologies and Tools

Backend	Node.js
Frontend	React, HTML, CSS, JavaScript
Database	MySQL / PostgreSQL / Google Cloud SQL
Version Control	GitHub
Prototype	Figma

1.6 Description of the finished product, service or result

The Cloud-Based Pharmacy Management System will support three user roles: Pharmacist, Doctor and Admin, each of them with secure login access. Pharmacists will be able to check medicine availability and expiry dates, search patients, view prescriptions, complete prescriptions and receive alerts for low or expiring stock. Doctors will be able to manage patient records and issue or update electronic prescriptions, which will become immediately available to pharmacists. Administrators will manage inventory, control user roles and permissions, generate reports on sales and stock and synchronize data across all pharmacy branches to keep information accurate and up to date.

1.7 Project Scope and Limitations

The following limitations have been identified and must be considered throughout the project lifecycle:

- Integration with national e-prescription systems and doctors' portals may require external approvals that are outside the team's control.
- Data synchronization must remain stable even when a pharmacy branch is temporarily offline.
- Role-based access control and full auditability are required to meet data privacy standards.

1.8 Project Schedule and Milestones

Phase / Milestone	Date
Requirements Analysis	March 2026
System Design	March – April 2026
Approved Requirements Milestone	28 March 2026
Prototype Ready Milestone	24 April 2026
Implementation	April – May 2026
Feature-Complete System Milestone	29 May 2026
Testing and Documentation	June 2026
Final Version Submitted	30 June 2026

1.9 Project Team

Project Manager	Andrea Kitanovska (89231153)
Backend Developer	Samoil Gjorgiev (89231015)
Frontend Developer & Documentation	Ana Milenkoska (89231022)
Frontend Developer & Documentation	Hristijan Kochovski (89231045)

1.10 Risk Assessment

The following risks have been identified as the most significant threats to successful project delivery:

- Data migration: Moving existing patient records, prescriptions, inventory, and supplier data into the new system poses a high risk due to potential inconsistency, incompleteness or loss of legacy data.
- Changing requirements: Client requirements may evolve during development, affecting scope and timeline.
- Integration challenges: Connecting frontend with backend and external healthcare systems may introduce unforeseen technical difficulties.
- Security and performance: Handling sensitive patient data requires robust security measures and reliable system performance under load.

Conclusion:

The Project Charter establishes a clear and shared understanding of the project's vision, scope, team responsibilities and constraints. It forms the contractual and organizational foundation upon which all subsequent development activities are built.

Chapter 2: Personas and Problem Scenarios

Understanding the end users of a system is essential for designing software that is both useful and usable. This chapter presents three user personas derived from research into the primary stakeholders of the pharmacy management system. For each persona, two key problem scenarios are described, along with the current limitations and the value proposition offered by the proposed system.

2.1 Persona 1: Dr. Vangel Markovski — General Practitioner

2.1.1 Profile

Age	41
Role	General Practitioner at a public hospital
Experience	15+ years in clinical practice
Daily workload	25–35 patients per day
Current setup	Partially digital hospital system; occasional use of paper prescriptions

Dr. Vangel is a doctor who relies on fast and accurate access to patient information. He becomes frustrated when systems are slow, incomplete, or fragmented and is deeply concerned about the risks of prescribing medication that could cause allergic reactions or interact with existing treatments. His primary goal is to treat patients efficiently while minimizing medical errors.

Thinks

- “If I make a mistake in prescribing medication, it could seriously harm someone.”
- “The system should help me, not slow me down.”
- “Everything should be in one place.”

Sees

- Fragmented patient records across systems.
- Paper prescriptions being used in some cases.
- Increasing digitalization in healthcare.

Feels

- Time pressure during peak hours.
- Concern about patient safety.
- Frustration when systems are slow or crash.
- Relief when processes work smoothly and reliably.

Does

- Logs into the hospital system daily.
- Issues multiple prescriptions per shift.
- Updates diagnoses and treatment plans.

2.1.2 Problem Scenario A: Adding a New Patient

When a new patient requests to register with Dr. Vangel as their general practitioner, he must add the patient to the centralized database, verify that no duplicate record exists in the national system, and ensure all data is accurate and immediately available for future consultations. The current fragmented system increases the risk of duplicate entries and delays treatment initiation.

Current limitations: The local hospital system is not synchronized with other institutions. Paper registration forms must be manually entered into a computer, and duplicate records are difficult to detect without a centralized validation mechanism.

Value proposition: The CBPMS provides a cloud-based centralized database with real-time duplicate detection and input validation, ensuring that patient records are accurate, complete, and instantly accessible across all authorized locations.

2.1.3 Problem Scenario B: Searching for an Existing Patient

When a patient arrives for an examination, Dr. Vangel must quickly retrieve the patient's complete medical history, previous diagnoses, known allergies, and active prescriptions. Under the current system, this often requires consulting multiple local databases or contacting other institutions, leading to delays and incomplete information.

Current limitations: The local-only database does not reflect data from other institutions. Paper files may need to be consulted if the digital record is incomplete, and external calls may be necessary to retrieve full patient history.

Value proposition: With the CBPMS, a doctor can search using a unique patient ID and instantly retrieve a complete, synchronized medical record, regardless of where the patient was previously treated.

2.2 Persona 2: Pharmacist Lea Zajkov — Private Pharmacy Chain

2.2.1 Profile

Age	35
Role	Pharmacist at a private pharmacy chain
Experience	11 years in pharmacy and medication management
Daily workload	High volume of prescription processing per shift
Current setup	Older-style system with inconsistent data and limited branch synchronization

Lea is a highly experienced pharmacist who takes patient safety very seriously. She is frustrated by systems that require excessive manual effort and that do not provide reliable, up-to-date prescription information. Her goal is to dispense medication accurately and efficiently, even during peak hours.

Thinks

- "If I miss a prescription detail, it could make the patient's health worse"
- "The system should provide the correct information."
- "Prescription status for particular patient and the stock levels should update automatically"

Sees

- Patients' expectations are fast service and immediate access to their prescription.
- Different branches happen to have different information.
- A lot of manual steps, i.e. retyping prescription, checking shelves for stock or inputting a patient in the system.

Feels

- Pressure during busy hours when the number of patients is bigger than usual.
- Confident when everything works 'by the book'

Does

- Logs into the pharmacy system if she is the first on her shift.
- Issues multiple prescriptions per shift.
- Searches for patients by unique ID to open their medical record.
- Checks allergies, conditions, and prescription details before dispensing.
- Marks prescriptions as completed so stock is updated automatically.
- Opens the alerts section to check low stock and soon-to-expire medicines.

2.2.2 Problem Scenario A: Searching for a Patient to Dispense Medication

When a patient arrives to collect medication, Lea searches for the patient by unique ID, reviews their prescription details and allergies, and dispenses the correct medication. After dispensing, she marks the prescription as completed so that stock levels are updated automatically. If the patient is not found, the system must display a clear error message.

Current limitations: The local desktop system is not connected to other branches, and prescription information is often retyped manually from paper or external platforms, introducing delays and potential errors.

Value proposition: The CBPMS enables real-time patient search, instant prescription retrieval, and automatic stock updates upon prescription completion, eliminating manual steps and reducing error risk.

2.2.3 Problem Scenario B: Handling a Prescription Update from a Doctor

A patient arrives to collect a prescription, but the prescribing doctor has updated the dosage and added a new medication earlier that day. Lea must confirm whether the system reflects the latest version before dispensing. Under the current setup, this requires calling the doctor's office or manually comparing paper records with the system's data, a process that is slow and error-prone.

Current limitations: The local system does not synchronize automatically with updates made in the doctor's portal. Manual verification is required, consuming time and increasing the risk of dispensing an outdated prescription.

Value proposition: The CBPMS synchronizes prescription updates from doctors in real time and notifies the pharmacist if a prescription has been modified since the patient's last visit, enabling safe and efficient dispensing without manual cross-checking.

2.3 Persona 3: Elena Petrova — Foreign University Student

2.3.1 Profile

Age	22
Role	International Computer Science student studying abroad
Medical history	Stored only in home country's healthcare system
Current challenge	Must re-register and recall medical history from memory at each new healthcare provider

Elena is a young student who faces significant challenges when seeking medical care abroad. She worries about forgetting to mention allergies or past prescriptions, and is frustrated by the need to re-register and repeat her medical history every time she visits a new doctor. Her primary concern is receiving safe and accurate treatment in an unfamiliar healthcare environment.

Thinks

- "I hope they can see my medical history."
- "What if I forget to mention an allergy?"
- "Why do I need to register again every time?"
- "My data should already be in the system."

Sees

- Different healthcare systems in different countries.
- Manual registration forms.
- Doctors asking for repeated information.
- No connection between her home and host country systems.

Feels

- Anxious when visiting a new doctor.
- Frustrated by repeated paperwork.
- Concerned about possible medical mistakes.
- Relieved when healthcare staff have access to accurate information.

Does

- Provides personal information repeatedly.
- Explains her medical history from memory.
- Keeps photos of prescriptions on her phone.
- Contacts her family doctor back home for confirmation.

2.3.2 Problem Scenario A: Seeking Medical Attention Abroad

Elena becomes ill while studying abroad and visits a local hospital. The attending doctor needs access to her medical history, allergies, and active prescriptions, but this information is stored only in her home country's system. She must provide her medical background manually, risking incomplete or inaccurate information that could delay or compromise her treatment.

Current limitations: Elena must register as a completely new patient, provide her medical history from memory, or wait for records to be sent from her home country — all of which introduce delays and safety risks.

Value proposition: The CBPMS stores patient records in a centralized cloud database accessible to authorized doctors across locations. Using a unique patient ID, any authorized

healthcare provider can instantly access Elena's full medical history, eliminating repeated registration and improving treatment safety.

2.3.3 Problem Scenario B: Emergency Care While Traveling

Elena suffers a severe allergic reaction while attending a university event. Emergency medical staff need immediate access to her allergy history, current medications, and chronic conditions. Without a centralized system, the staff have no access to this critical information, and Elena is unable to recall all relevant details accurately under stress.

Current limitations: Medical staff must rely on information provided verbally by the patient, which may be incomplete or inaccurate in an emergency situation. Contacting the home-country doctor takes valuable time.

Value proposition: The CBPMS makes critical patient information securely available in real time to authorized emergency staff anywhere, significantly reducing the risk of medical errors in time-sensitive situations.

Conclusion:

The three personas identified in this chapter — a general practitioner, a pharmacist, and a foreign student — represent the primary user groups of the system. Their problem scenarios reveal clear and consistent pain points related to data fragmentation, manual processes, and lack of real-time synchronization. These insights form the basis for the user stories and functional requirements developed in the following chapter.

Chapter 3: User Stories and Acceptance Criteria

User stories translate the difficulties identified through persona research into concrete, testable requirements for the system. Acceptance criteria define the conditions that must be satisfied for each story to be considered complete.

3.1 Doctor User Stories

3.1.1 Patient Search

As a doctor, I want to search for a patient using their unique ID so that I can quickly access their medical history and treatment information during an examination.

- The system allows search by valid unique patient ID.
- The correct patient record, including medical history, allergies, diagnoses and active prescriptions, is retrieved.
- The record reflects the most recent updates.
- A "No records found" message is displayed for invalid or unknown IDs.

3.1.2 Digital Prescription Issuance

As a doctor, I want to issue a prescription digitally so that the patient can receive the correct medication and pharmacists can access the prescription instantly.

- A new prescription can be created directly from the patient's medical record.
- The system displays allergy warnings if the prescribed medication conflicts with known patient allergies.
- The prescription is saved, linked to the patient's record and immediately visible to authorized pharmacies.

3.1.3 Updating Diagnosis and Treatment Plan

As a doctor, I want to update a patient's diagnosis and treatment plan so that the medical record always reflects the patient's current health condition.

- The patient's medical record can be opened and updated with new diagnosis and treatment details.
- The system records the date, time and identity of the doctor making the change.
- Updated information is immediately visible to all authorized healthcare professionals.

3.2 Pharmacist User Stories

3.2.1 Patient Search and Prescription Access

As a pharmacist, I want to search for a patient using their unique ID and open the prescription record so I can dispense the right medicine to the right person.

- A patient can be searched by a unique ID and the record is returned within a few seconds.
- The record displays allergies, conditions, and full prescription details.
- The data reflects the latest updates from any branch or doctor.
- An appropriate error message is shown for missing or invalid IDs.

3.2.2 Prescription Completion and Stock Update

As a pharmacist, I want to view the latest version of a prescription and complete dispensing in one flow so that I do not dispense wrong medication or an incorrect dose.

- The latest prescription version, including any recent changes made by the doctor, is displayed before dispensing.
- The timestamp of the last prescription update is clearly shown.
- Upon completion, the prescription status is set to "Completed", stock is updated, and the system prevents duplicate fulfillment.

3.2.3 Stock and Expiry Alerts

As a pharmacist, I want to receive alerts about low-stock and near-expiry medicines so I can prevent situations where patients are affected by shortages or expired medications.

- The "View Alerts" panel displays all medicines below the minimum stock threshold.
- The panel also shows all medicines approaching their expiry date.
- The message "No alerts at this time" is shown when no issues are detected.
- The panel refreshes automatically after stock updates to prevent stale alert data.

3.3 Patient - International Student User Stories

3.3.1 Cross-Border Medical Record Access

As a foreign university student, I want doctors abroad to access my medical history using my unique ID so that I can receive safe treatment without repeating my medical information.

- Doctors can search patients using a unique ID and retrieve full medical history, allergies, and prescriptions.
- Information is always up to date.
- A clear "No records found" message is displayed if the ID does not exist.

3.3.2 Emergency Information Visibility

As a student experiencing a medical emergency abroad, I want emergency doctors to quickly view my allergy and medication information so that they can avoid medical mistakes.

- Allergy information is prominently displayed at the top of the patient record.
- Current medications are shown with dosage details.
- Critical medical information is visually highlighted for emergency use.

3.3.3 Avoiding Repeated Registration

As an international student, I want my medical data to be available in hospitals in different countries so that I do not need to register again every time I visit a doctor.

- Doctors in different hospitals and countries can access the same patient record.
- The system prevents duplicate patient registrations.
- Patient data is synchronized across all system locations.
- Prescription updates are visible regardless of where they were issued.

Conclusion:

The user stories defined in this chapter provide a clear, structured and testable specification of the system's expected behavior from the perspective of each primary user group. They serve as the primary input for development planning and quality assurance activities.

Chapter 4: Hypothesis Testing and User Interviews

This chapter presents three core hypotheses derived from the user personas, along with the findings from structured interviews conducted with representative users. The goal is to confirm that the identified problems are real, frequent and sufficiently significant to justify the proposed solution.

4.1 Hypothesis Testing

Hypothesis 1 — Doctors

Do doctors encounter problems frequently enough with scattered patient records and duplicate entries that they would prefer a single centralized system enabling quick patient lookup and access to complete medical history?

Hypothesis 2 — Pharmacists

Do pharmacists deal often enough with outdated prescriptions and manual stock checks that they would find real-time prescription updates, automatic stock management, and stock alerts genuinely beneficial?

Hypothesis 3 — Patients (International Students)

Do patients receiving care away from home face sufficient difficulties with repeated registration and reliance on memory for medical information that a centralized accessible system would significantly improve their experience?

4.2 Interview Findings

4.2.1 Doctor — General Practitioner at a Public Hospital

Q: Tell me about the last time you had to add a new patient to the system.

"A new patient came with a paper referral and some printed lab results. I entered the ID in the local system, but I still had to check the documents manually because I was not sure whether the patient already existed in another institution."

Q: How do you currently check whether a patient already exists?

"Usually I search the local hospital database first. If I do not find enough information, I ask the patient or check their prescription papers. If they have been treated somewhere else, I may not see that."

Q: What would a better system need to do?

"It has to be fast, show the full patient history in one place, validate the patient ID immediately, and clearly show when the record was last updated."

4.2.2 Pharmacist — Private Pharmacy

Q: What do you do when a patient arrives before dispensing medication?

"I find the patient in the system, but I still check the paper copy to make sure it is the same."

Q: How do you know you are looking at the latest prescription version?

"I check the system and verify whether the prescription data matches the data from the patient's last doctor's visit."

Q: How do you currently track low-stock and near-expiry medicines?

"We do it manually. Expiry dates are the easiest thing to miss when the pharmacy is busy."

Q: What would a better system need to do?

"It should show me the correct and latest prescription straight away, make allergies easy to notice and update stock automatically. If it works reliably when it is busy, it will help my work life."

4.2.3 Patient — Foreign University Student

Q: Tell me about the last time you needed medical care while abroad.

"I went to a clinic because of an infection. Before I saw the doctor, I had to answer questions in a foreign language, which was really uncomfortable given my condition."

Q: What risk worried you most?

"Forgetting an allergy or getting medicine that could conflict with something I already take."

Q: What would a better system need to do?

"A better system would mean less work for me, with all the information in one place and accessible simply by using a personal ID."

4.3 Key Findings and Identified Improvements

The interviews confirmed all three hypotheses. Doctors do struggle with fragmented records and duplicate patient entries. Pharmacists do encounter outdated prescriptions and manual stock management challenges. And patients receiving care abroad do face repeated registration and the burden of recalling critical medical details from memory.

In addition to confirming the core problem statements, the interviews revealed several specific improvements that users would find valuable:

- Displaying when a record or prescription was last updated and by whom.
- Clearly highlighting what has changed in an updated prescription.
- Placing key safety information — such as allergies and current medications — at the top of the patient record screen.
- Showing whether data is fully synchronized or still pending an update.

Conclusion: Hypothesis testing confirmed that the problems addressed by the CBPMS are real, frequent, and impactful for all three user groups. The interview insights also provided actionable guidance for refining the user interface, which informed subsequent usability testing and design improvements.

Chapter 5: Usability Testing

With the hypotheses validated and the prototype developed, the next step was to evaluate whether the system's core flows are intuitive and efficient in practice. This chapter presents the setup, participants, tasks and results of the usability testing sessions conducted on the CBPMS prototype.

5.1 Testing Type and Goal

The team conducted assessment usability testing to determine whether users can complete the main system flows without confusion or significant delays. The objectives were to evaluate whether primary users can complete core tasks with minimal guidance, to identify any parts of the interface that cause confusion and to compare perceived efficiency against the users' current systems.

The following functionalities were covered in testing: Add Patient, Search Patient / View Medical Record, Mark Prescription as Fulfilled and View Alerts.

5.2 Participants

Participant 1	A doctor at a public hospital — issues prescriptions regularly; uses a digital or partly digital system.
Participant 2	Pharmacist at a private pharmacy — dispenses prescription medicine regularly; uses a digital or partly digital system.
Participant 3	Pharmacist at a pharmacy chain — dispenses prescription medicine regularly; uses a digital or partly digital system.

5.3 Session Structure

Each session was structured to last approximately 40 minutes, divided as follows: a 5-minute introduction, a 15-minute discovery phase with background questions, a 15-minute task phase and a 5-minute debrief. Sessions began from the same starting page and participants were not directed where to click. Observations were recorded during and immediately after each session.

5.4 Tasks

Doctor Tasks

- Add a new patient using a unique ID.
- Search for an existing patient and open their medical record.
- Find which prescriptions are currently In Progress.

Pharmacist Tasks

- Search for a patient by unique ID.
- Check the prescription details and mark it as fulfilled.
- Open the Alerts panel and identify one low-stock and one near-expiry medicine.

5.5 Task Completion Times

Participant	Task 1Duration	Task 1Est.	Task 2Duration	Task 2Est.	Task 3Duration	Task 3Est.
Doctor	1 min	40 s	49 s	1.1 min	30 s	30 s
Pharmacist 1	58 s	35 s	1.3 min	45 s	1 s	10 s
Pharmacist 2	47 s	35 s	57 s	45 s	7 s	10 s

5.6 Key Findings and Recommended Improvements

All three participants successfully completed all assigned tasks, confirming that the core system flows are functional and understandable. The easiest task for all users was locating the low-stock alerts. The most challenging flow was checking a prescription and marking it as fulfilled, as it required first locating the patient before proceeding to the prescription.

The following improvements were identified for the next development iteration:

- Relocate certain functionalities to improve visibility and reduce navigation steps.
- Add clearer confirmation messages when actions such as adding a patient are completed successfully.
- Improve feedback during the Add Patient flow to reduce hesitation and uncertainty.
- Provide explicit visual indicators for prescription status and stock changes.

Conclusion:

The usability testing confirmed that the prototype is generally intuitive and supports task completion for all primary user roles. The identified issues are addressable through targeted interface improvements and do not affect the fundamental architecture or logic of the system.

Chapter 6: Agile Development — Scrum Sprints

This chapter summarizes the sprint structure, goals, scope, and outcomes for each phase of the project.

6.1 Sprint 0: Project Setup and Planning

Sprint Goal: Define project scope and structure by planning and organizing upcoming sprints.

Sprint 0 established the organizational foundation for the project. The team aligned on the overall vision, selected the technology stack, defined team roles and produced the sprint roadmap for the subsequent three sprints.

Key Deliverables

- Defined project scope and objectives.
- Sprint roadmap covering Sprints 1 through 3.
- Initial feature list identifying core system capabilities.
- Technology stack selection (React, Node.js, cloud database, GitHub).
- Team roles and workflow defined.

6.2 Sprint 1: Personas, Scenarios, and User Stories

Sprint Goal: Define clear personas, identify key problem scenarios and translate them into structured user stories with acceptance criteria.

Sprint 1 focused on user research and requirements definition. The team produced three fully developed personas, documented six problem scenarios and structured a comprehensive set of user stories with testable acceptance criteria. These artifacts formed the primary input for subsequent design and validation work.

Key Deliverables

- Three fully defined personas (Doctor, Pharmacist, International Student).
- Documented problem scenarios for each persona.
- Structured user stories following the standard format.
- Testable acceptance criteria for each user story.

6.3 Sprint 2: Validation and Usability Testing

Sprint Goal: Validate system assumptions through user feedback and improve usability of core features.

Sprint 2 focused on hypothesis testing and usability evaluation. All three hypotheses were validated through structured user interviews and a prototype usability test was conducted with three participants representing the primary user roles. The sprint produced a clear set of validated findings and actionable improvement recommendations.

Key Findings

- The identified problems are real and confirmed for all three user groups.
- The prototype is generally easy to use; all tasks were completed successfully.

- Main usability issues: Add Patient flow clarity, prescription completion complexity and insufficient system feedback.

6.4 Sprint 3: Prototype Improvement and System Implementation

Sprint Goal: Improve the prototype based on usability feedback and implement core functionalities with a real database.

Sprint 3 represents the core development phase of the project. It encompasses interface improvements derived from Sprint 2 findings, full implementation of the primary system features and integration with a centralized cloud database.

UI/UX Improvements

- Improved Add Patient flow clarity and error feedback.
- Simplified prescription completion flow.
- Added clear success and error confirmation messages.
- Highlighted allergies and critical patient data visually.
- Displayed last update timestamp and the author of changes.
- Added sync status indicators (Synced / Pending update).

Functional Development

- Patient registration with duplicate validation.
- Patient search by unique ID.
- Digital prescription creation and update functionality.
- Prescription status management (In Progress / Completed).
- Automatic stock updates after prescription fulfillment.
- Low-stock and near-expiry alert system.

Database Integration

- Centralized database design and connection.
- Storage of patient records, prescriptions, and inventory data.
- Basic real-time data synchronization across branches.

Conclusion:

The Scrum sprint structure ensured that the project progressed in a controlled, iterative manner. Each sprint built directly on the outputs of the previous one, from research and requirements through validation and into implementation. This approach allowed the team to incorporate user feedback continuously and deliver a system that genuinely addresses the needs of its target users.

Chapter 7: Project Stakeholder Analysis

A thorough understanding of project stakeholders is essential for managing expectations, addressing potential conflicts and ensuring that the system delivers value to all parties involved. This chapter identifies the four primary stakeholder groups for our system, describes their interests and concerns and outlines the team's approach to managing each relationship.

7.1 Pharmacists

Status: Positive

Expectations and interests: Pharmacists expect fast patient search, clear and up-to-date prescription details, automatic stock updates and timely alerts for low stock and near-expiry medicines. They want to minimize manual effort and reduce the risk of dispensing errors.

Potential conflicts: If the system is slow, unclear or fails to reflect the latest prescription changes in real time, pharmacists may lose confidence in it and revert to manual workarounds.

Team response: The pharmacist interface will prioritize simplicity, clear visual hierarchy, real-time updates and explicit confirmation messages for all critical actions.

7.2 Suppliers and Medicine Distributors

Status: Neutral to positive

Expectations and interests: Suppliers require accurate and timely visibility of stock levels, automatic notifications when inventory falls below thresholds and smooth coordination with pharmacy ordering processes.

Potential conflicts: Inaccurate or delayed stock data could lead to missed orders, shortages or overstocking, which would damage the business relationship.

Team response: The team will ensure reliable and consistent stock tracking, simulate supplier-facing data flows before going live, and design the system to support future API-based integration with supplier platforms.

7.3 Health Authorities — Ministry of Health

Status: Neutral

Expectations and interests: The Ministry of Health expects the system to handle sensitive patient data securely, enforce strict access controls, maintain full audit trails and comply with applicable national healthcare regulations and data privacy standards.

Potential conflicts: Any failure in data security, privacy compliance or auditability could result in regulatory sanctions and reputational damage.

Team response: Security measures — including role-based access control, data encryption, and comprehensive audit logging — will be integrated from the earliest stages of development. Only essential data will be stored to minimize privacy exposure.

7.4 Patients

Status: Positive

Expectations and interests: Patients expect accurate prescriptions, clear medication instructions, reliable service, and confidence that their personal and medical data is handled with the highest level of privacy and security.

Potential conflicts: Prescription errors, service delays or data privacy concerns could erode patient trust in the system and in the healthcare providers who use it.

Team response: The team will implement prescription validation, clear and readable patient-facing information and robust data protection measures to maintain patient trust and safety.

Conclusion:

Engaging stakeholders proactively and understanding their distinct interests is essential for the long-term success of the CBPMS. By addressing the needs and concerns of pharmacists, suppliers, health authorities, and patients from the outset, the team can build a system that is not only technically sound but also trusted and valued by all parties who interact with it.

Chapter 8: Engineering Management — Patterns and Practices

Successful software engineering requires not only technical competence but also engineering management skills. This chapter examines two negative patterns that pose risks to the project and two positive patterns that the team has committed to implementing. Recognizing these patterns early and responding to them proactively is a key element of professional engineering management.

8.1 Negative Pattern: Unclear Ownership of Critical Work

Description: Unclear Ownership of Critical Work is a negative pattern where important project responsibilities are not clearly assigned to specific team members. As a result, critical tasks may be ignored, duplicated, or completed inconsistently.

Risks in This Project

In the CBPMS context, this pattern could manifest in the following ways:

- Duplicate work caused by multiple team members unknowingly working on the same feature.
- Missing integration between frontend and backend components.
- Delays in implementing critical healthcare workflows such as prescription processing or stock synchronization.
- Confusion about who is responsible for fixing bugs or completing unfinished tasks.
- Incomplete documentation, testing, or presentation preparation.

The consequences of unclear ownership are especially serious in healthcare systems because disconnected or unfinished features can reduce reliability and create risks in patient-related operations.

Mitigation Strategy

- Clearly assign responsibility for each major feature and workflow.
- Define ownership for backend logic, frontend interfaces, testing, and documentation.
- Track tasks and progress regularly through team meetings or project management tools.
- Ensure every feature has one main responsible member, even when collaboration is involved.
- Review integration between connected modules frequently during development.

8.2 Negative Pattern: Last-Minute Development

Description: The "Last-Minute Development" pattern occurs when teams defer important tasks to the final stages of the project, resulting in rushed implementation, inadequate testing and reduced quality.

Risks in This Project

- Critical features implemented too late to be properly tested or refined.
- Documentation written under pressure immediately before submission.
- Bugs discovered too close to the deadline to be resolved effectively.
- Increased team stress and reduced overall work quality.

Mitigation Strategy

- Divide the project into structured sprints with clear weekly goals and milestones.
- Monitor progress continuously through GitHub commits and regular team meetings.
- Test features incrementally throughout development rather than at the end.
- Maintain up-to-date documentation as part of each sprint's deliverables.

8.3 Positive Pattern: Continuous Integration and Testing

Description: Continuous Integration and Testing is the practice of regularly merging code changes into a shared repository, followed by automated or manual testing after each integration. This ensures that problems are detected and resolved early.

Application in This Project

- All code changes are uploaded to GitHub and reviewed before merging.
- New features are tested after every update to verify correct behavior.
- Core modules, prescription management, inventory tracking, patient records, are validated continuously.
- Bugs are addressed immediately upon discovery.

This pattern is especially critical in healthcare software, where a single undetected defect can have direct consequences for patient safety.

8.4 Positive Pattern: Clear Feature Ownership

Description: Clear Feature Ownership is the practice of assigning responsibility for important system features to specific team members. This does not mean that only one person works on a feature, but rather that one person ensures the feature is completed correctly and integrated properly with the rest of the system.

Application in This Project

- All major features are assigned to responsible team members based on their roles.
- Backend-related tasks such as database management and system logic are handled by backend developers.
- Frontend members manage user interfaces for doctors, pharmacists, and administrators.
- Documentation and reporting responsibilities are clearly divided among team members.
- Project progress is monitored through defined ownership of patient management, prescription processing, inventory tracking, alerts and role-based access features.

This pattern is especially important in healthcare software projects, where unclear responsibilities can lead to missing functionality, communication problems, or incomplete workflows that affect system reliability.

Conclusion:

Clear feature ownership is very useful for our project because the system includes several user roles and connected workflows. When ownership is clear, the team can build the system in a more organized and reliable way.

Conclusions

The Cloud-Based Pharmacy Management System project has progressed methodically through each phase of the software engineering lifecycle, from initial problem definition and user research through hypothesis validation, prototype development, usability testing and agile implementation.

The research conducted through personas, problem scenarios, interviews and usability testing consistently confirmed the relevance and urgency of the problem being addressed. Doctors are hindered by fragmented patient records, pharmacists are burdened by manual processes and outdated information and patients - particularly those seeking care outside their home country, face unnecessary barriers and safety risks.

The proposed system directly addresses each of these challenges through centralized cloud-based storage, real-time data synchronization, role-based access control and an interface designed to support fast and safe decision-making in clinical environments.

As the project moves toward its final delivery milestone, the team is focused on completing the functional implementation, finalizing database integration and ensuring that all usability improvements identified during testing are incorporated into the production system.

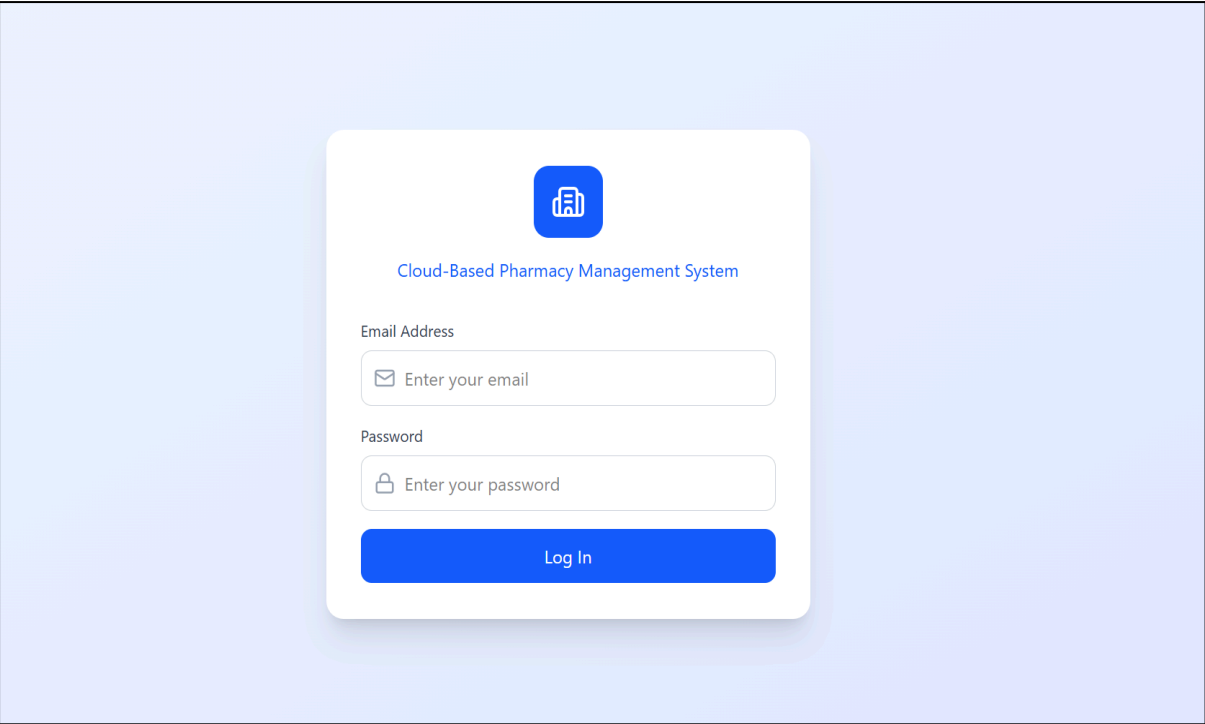
The Cloud-Based Pharmacy Management System represents not only a technical achievement but a meaningful contribution to patient safety, operational efficiency and the modernization of pharmacy services within the national healthcare infrastructure.

Work assessment:

For the homework and documentation part, all team members contributed equally to the research, writing, testing, planning and final documentation.

For the system development, everyone contributed equally while working according to their assigned roles: Andrea as project manager, Samoil on backend development, Ana and Hristijan on frontend development and documentation.

Screenshots of the final application:



Cloud-based Pharmacy Management System

Dashboard

Search Inventory

Search Patient

Dashboard

3,847

Items in Stock

23

Expiring Soon (30 days)

Low Stock Alerts

Amoxicillin 500mg

45/100

low

Lisinopril 10mg

22/50

critical

Metformin 850mg

78/100

low

Atorvastatin 20mg

15/75

critical

[View Full Inventory](#)

Items Expiring Soon

Product Name	Batch Number	Expiry Date	Days Remaining
Ibuprofen 400mg	BT-2024-A45	2026-12-15	314 days
Paracetamol 500mg	BT-2024-B12	2027-12-28	427 days
Cetirizine 10mg	BT-2024-C89	2028-01-05	535 days

Cloud-Based Pharmacy Management System

Cloud-based Pharmacy Management System

Dashboard

Search Inventory

Search Patient

Search Inventory

Barcode Import

Add Item

Search by name or SKU...

All Categories

More Filters

Total Items6

In Stock2

Low Stock2

Expiring Soon1

Product	Category	Quantity	Batch	Expiry	Status
Amoxicillin 500mg PharmaCorp Inc.	Antibiotics	450 Min: 100	BT-2024-A123	2026-06-15	in stock
Lisinopril 10mg MediSupply Co.	Cardiovascular	22 Min: 50	BT-2024-B456	2026-08-20	low stock
Metformin 850mg HealthMeds Ltd.	Diabetes	0 Min: 100	BT-2024-C789	2026-04-10	out of stock
Ibuprofen 400mg PharmaCorp Inc.	Pain Relief	280 Min: 150	BT-2024-D012	2028-12-15	expiring soon
Atorvastatin 20mg MediSupply Co.	Cardiovascular	15 Min: 75	BT-2024-E345	2027-09-30	low stock
Omeprazole 40mg HealthMeds Ltd.	Gastrointestinal	320 Min: 100	BT-2024-F678	2026-07-25	in stock

Cloud-based Pharmacy Management System

Dashboard

Search Inventory

Search Patient

Patient Records

Access patient medical history and prescription records

Search by patient ID

Cloud-based Pharmacy Management System

Dashboard

Search Inventory

Search Patient

Patient Records

Access patient medical history and prescription records

45678

Novak Janez

ID: PT-45678 Age: 40 DOB: 1985-03-15

+1 (555) 123-4567 novakjanez@email.com

Allergies: Penicillin, Sulfa drugs

View Full Record

Cloud-based Pharmacy Management System

Dashboard

Manage Inventory

Report sales, stocks, suppliers

Synchronize pharmacy branches

Manage roles

Manage roles

Total Users

3

Active Users

3

Admins

1

Pharmacists

1

Search by name, email, or branch...

All Roles

User	Role	Status	Last Login	Actions
Admin User admin@cloudpharm.com	admin	active	Dec 1, 11:30 AM	
Sarah Johnson sarah.johnson@cloudpharm.com	pharmacist	active	Dec 1, 10:15 AM	
Michael Chen michael.chen@cloudpharm.com	doctor	active	Dec 1, 09:45 AM	

Cloud-based Pharmacy Management System

Dashboard

Manage Inventory

Report sales, stocks, suppliers

Synchronize pharmacy branches

Manage roles

Synchronize pharmacy branches

My Branch

BR-001

active

Goriska ulica 23

+386 (55) 100-0001

Manager: Jasna Skufca

Employees

Inventory

Sales (Month)

12

3,847

\$145.89k

Last sync: 01/12/2024

View Details

Pharmacy Titov Trg

BR-002

active

Titov Trg ST 12346

+386 (54) 100-0002

Manager: Patrik Boncin

Employees

Inventory

Sales (Month)

8

2,156

\$98.45k

Last sync: 01/12/2024

View Details

Pharmacy Taverna

BR-003

active

Kidriceva ulica 12

+386 (45) 100-0003

Manager: Emilija Davidovska

Employees

Inventory

Sales (Month)

6

1,823

\$76.23k

Last sync: 01/12/2024

View Details

Pharmacy Planet Koper

BR-004

active

Berneticeva ulica 12

+386 (35) 100-0004

Manager: Robert Moris

Employees

Inventory

Sales (Month)

10

2,943

\$112.34k

Last sync: 01/12/2024

View Details

Real-Time Synchronization Status

My Branch

Database synchronized

01/12/2024

All systems operational

Pharmacy Titov Trg

Database synchronized

01/12/2024

All systems operational

Pharmacy Taverna

Database synchronized

01/12/2024

All systems operational

Pharmacy Planet Koper

Database synchronized

01/12/2024

All systems operational

28

Patient Medical Record

Novak Janez (PT-45678)

Personal Information

Full Name	Patient ID
Novak Janez	PT-45678
Date of Birth	Phone
1985-03-15 (40 years old)	+1 (555) 123-4567
Email	Address
novakjanez@email.com	Prisoje 9, Koper 6000

Medical Information

Allergies	Medical Conditions
Penicillin Sulfa drugs	Hypertension Type 2 Diabetes

Prescription

RX-2024-1234 Pending

📅 2024-12-01 • Dr. Sarah Smith

Medications:

📄 Amoxicillin 500mg

📄 Ibuprofen 400mg

Close Mark Prescription as COMPLETED

Patient Medical Record

Novak Janez (PT-45678)

Personal Information

Full Name	Patient ID
Novak Janez	PT-45678
Date of Birth	Phone
1985-03-15 (40 years old)	+1 (555) 123-4567
Email	Address
novakjanez@email.com	Prisoje 9, Koper 6000

Medical Information

Allergies	Medical Conditions
Penicillin Sulfa drugs	Hypertension Type 2 Diabetes

Prescription

RX-2024-1234 completed

📅 2024-12-01 • Dr. Sarah Smith

Medications:

📄 Amoxicillin 500mg

📄 Ibuprofen 400mg

Close COMPLETED